

One CSI Report Cannot Serve Every Consumer: The Consumer-Relative False-Clear Rate in 5G/6G Link Adaptation

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Abstract—A channel-state-information (CSI) report is a certificate: an implicit claim that the channel is good enough to transmit at a selected rate. That certificate is not read by the channel; it is read by a *consumer*—a scheduler with a reliability target and a decoder at a moment in time. We make the rate at which a CSI certificate *false-clears* (selects a rate the link cannot support) a first-class, measured KPI on a real 5G NR link-level chain (Sionna LDPC decoding), and show it is *consumer-relative* along two axes. First, the reliability target: the identical CSI-derived modulation-and-coding selection that meets an eMBB target (BLER = 10^{-1}) misses a URLLC target (10^{-3}) by more than two orders of magnitude ($112\times$), while a target-aware policy meets it. Second, the refresh horizon: as the report ages, the false-clear rate grows, and the largest safe report period is proportional to the Clarke coherence time ($\approx 0.177 T_{\text{coh}}$, $R^2 = 0.915$). The witnessed corrections that hold each axis—outer-loop link adaptation, a coherence-driven refresh—already exist; our contribution is the measured number that says when each is needed. All results are drawn from sealed, pre-registered records.

Index Terms—CSI feedback, link adaptation, URLLC, outer-loop link adaptation, reliability, 5G NR.

I. INTRODUCTION

Adaptive modulation and coding (AMC) turns a channel-state-information (CSI) report into a rate decision: the transmitter selects the highest modulation-and-coding scheme (MCS) whose required SINR the reported CSI clears. The report is thus a *certificate*—an implicit assertion that the link will decode at the chosen rate. Like any certificate, it can be wrong in a way that matters only to whoever reads it. We call a wrong-in-the-optimistic-direction certificate a *false-clear*: the CSI cleared an MCS that the true channel cannot support, and the transport fails (a HARQ NACK, a block error).

Our thesis is that the false-clear rate is not a property of the channel or even of the report alone; it is a property of the report *and its consumer*. Two schedulers reading the same CSI can disagree on whether it false-clears, because they read it against different requirements. We make the false-clear rate a measured KPI on a real 5G NR link-level chain and demonstrate this consumer-relativity along two operationally important axes:

- **Reliability target** (§III): one CSI-derived MCS adequate for enhanced mobile broadband (eMBB) is catastrophically

cally optimistic for ultra-reliable low-latency communication (URLLC)—a false-clear by more than $100\times$.

- **Refresh horizon** (§IV): as the report ages the false-clear rate grows, and the safe refresh period scales with the channel coherence time by a simple measured law.

Neither observation requires a new estimator or scheduler. The corrections are known: outer-loop link adaptation (OLLA) [1] closes the target-relativity gap, and coherence-driven CSI reporting closes the refresh gap. What has been missing is the *measured, consumer-relative false-clear rate* that quantifies when each correction is load-bearing and how large the exposure is when it is absent.

II. METHOD

Substrate. We use the real 5G NR LDPC decoder in Sionna [2] as the error witness. For each MCS we first *measure* a block-error curve $e_m(\gamma)$ by decoding 400 coded blocks per SNR point (grid -8 to 28 dB, 2 dB spacing) over AWGN. The time-domain link then drives a per-TTI SNR trace through a 3GPP TR38.901 TDL-A channel (carrier 3.5 GHz, delay spread 30 ns) and, at each TTI, resolves the HARQ outcome of the selected MCS from e_m at the instantaneous SNR—a standard link-to-system mapping built on *measured* LDPC curves, not a fabricated formula. The MCS set is the 3GPP NR table (TS 38.214 Table 5.1.3.1-1; QPSK-64-QAM, code rates $R \geq 1/5$). Table I lists the parameters.

System model. Let $\gamma(t)$ (dB) be the true SNR at TTI t and $\hat{\gamma}(t)$ the CSI-reported estimate: for *fresh* CSI $\hat{\gamma}(t) = \gamma(t) + n(t)$, $n \sim \mathcal{N}(0, \sigma^2)$ with $\sigma = 1$ dB; for a report *held* with period P , $\hat{\gamma}(t) = \gamma(P\lfloor t/P \rfloor)$, so the estimate ages. Let $e_m(\gamma)$ be the measured LDPC error probability of MCS m and $\rho_m(\beta) = \min\{\gamma : e_m(\gamma) \leq \beta\}$ its required SNR at target β . The AMC *certificate* selects, using a *fixed reference* target $\beta_0 = 10^{-1}$ and backoff Δ ,

$$m^*(\hat{\gamma}) = \max\{m \in \mathcal{M} : \rho_m(\beta_0) + \Delta \leq \hat{\gamma}\}, \quad \Delta=0 \text{ (naive)}, \quad (1)$$

so the *operating point* is set by β_0 but the outcome is *judged* by the consumer's target β . The *consumer* is β ; the *witness* is the HARQ outcome $\text{NACK}(t) \sim \text{Bernoulli}(e_{m^*}(\gamma(t)))$. A false-clear is a cleared MCS the channel cannot support at β ,

TABLE I
LINK-LEVEL PARAMETERS (SEALED CELLS)

Parameter	Value
Carrier frequency	3.5 GHz
TTI duration	1 ms
Channel model	3GPP TR38.901 TDL-A, DS = 30 ns
Mean SNR	12 dB
Decoder / code	Sionna 5G NR LDPC (real decode)
MCS set	NR TS 38.214 Tbl 5.1.3.1-1 (QPSK-64QAM)
BLER curves	400 blocks/pt, SNR -8-28 dB @ 2 dB
CQI report noise σ	1 dB
Trace length	6000 TTI
eMBB / URLLC target	$10^{-1} / 10^{-3}$
URLLC-aware policy	conservative MCS + $K=3$ diversity
OLLA step $\delta_{\uparrow}$	0.1 dB ($\delta_{\downarrow}=0.9$ dB)
Main aging Doppler	200 Hz ($T_{\text{coh}} \approx 2.1$ ms, ≈ 62 km/h)
Refresh sweep f_D	10-400 Hz

i.e. $\rho_{m^*}(\beta) > \gamma(t)$. We report two estimators of its rate. For the refresh study (§IV) the *empirical* HARQ NACK rate

$$\widehat{\text{FC}} = \frac{1}{T} \sum_{t=1}^T \mathbf{1}[\text{NACK}(t)], \quad (2)$$

which the outer loop consumes; for the reliability-target study (§III), whose 10^{-3} tail $T=6000$ TTI cannot resolve empirically, the *expected* rate $\text{FC}(\beta) = \frac{1}{T} \sum_t e_{m^*}(\gamma(t))$, read against β . Because (1) fixes the operating point at β_0 while the verdict uses β , one trace $\{\hat{\gamma}(t), \gamma(t)\}$ yields different results for the eMBB ($\beta=10^{-1}$) and URLLC ($\beta=10^{-3}$) consumers—the consumer-relativity we measure.

URLLC-aware policy. It raises Δ (a conservative MCS) and combines $K=3$ diversity branches: a backoff alone cannot reach 10^{-3} , since a single Rayleigh branch has a $\sim 4\%$ deep-fade outage floor regardless of Δ . The branches $\{\gamma_k(t)\}$ are K independent TDL-A realizations (distinct seeds; frequency/spatial replicas, not retransmissions), and a block fails only if all fail, $\text{FC}_{\text{aware}} = \frac{1}{T} \sum_t \prod_{k=1}^K e_{m^*}(\gamma_k(t))$. This is a *best case*: independent branches lower-bound the joint outage, so correlated diversity raises FC_{aware} and the true cost of URLLC safety is at least what we report.

OLLA. The witnessed correction (§IV) distrusts $\hat{\gamma}$ and selects $m^*(\hat{\gamma}(t) - o(t))$, updating the offset asymmetrically from HARQ feedback,

$$o(t+1) = o(t) + \delta_{\uparrow} \mathbf{1}[\text{ACK}(t)] - \delta_{\downarrow} \mathbf{1}[\text{NACK}(t)], \quad (3)$$

with up-step $\delta_{\uparrow} = 0.1$ dB and $\delta_{\downarrow} = \delta_{\uparrow}(1 - \beta)/\beta$, so the equilibrium NACK rate is β . For held CSI the *refresh floor* is $P^*(f_D) = \max\{P : \widehat{\text{FC}} \leq \beta\}$.

Discipline. Each experiment is pre-registered with numerical bars and manipulation checks fixed before the graded run; the sealed record is content-hashed and graded on seeds disjoint from those used to design it, and negative or failing outcomes are kept [7]. The numbers below are the graded, sealed values (Table II).

III. CONSUMER-RELATIVITY IN THE RELIABILITY TARGET

A base station serving mixed traffic derives one CSI estimate per user and selects an MCS. Consider the same estimate

consumed under two targets: eMBB ($\text{BLER} \leq 10^{-1}$) and URLLC ($\leq 10^{-3}$). We drive a real NR link at high mobility, apply the naive CSI-derived MCS, and measure the achieved BLER against each target; we then compare a *target-aware* policy that sizes its margin to the consumer's target.

Figure 2 reports the sealed result (three graded seeds, $T=6000$ TTI, fresh CSI with $\sigma=1$ dB report noise; expected-rate estimator). The naive certificate achieves $\text{BLER} \approx 0.112$. Read by the eMBB consumer this is at target—the certificate is sound. Read by the URLLC consumer the *same* 0.112 is a false-clear by $112\times$ (against 10^{-3}): every such block is a reliability-SLA violation, invisible to a scheduler that assumed one CSI report serves all traffic. The target-aware policy—the same CSI, a conservative MCS backoff combined with the $K=3$ independent diversity branches of Sec. II (a backoff alone cannot reach the 10^{-3} tail on one branch)—achieves $\text{BLER} \approx 1.2 \times 10^{-5}$, comfortably within target, at the cost of the backoff's spectral efficiency and the diversity's resource overhead. No single certificate is *efficient* for both: the one that is sound for eMBB false-clears for URLLC by $112\times$, and the one that is sound for URLLC over-provisions eMBB. The false-clear rate is the number that prices the difference.

The relativity is not an artifact of the operating point. Sweeping the mean SNR from cell edge to cell centre (3–21 dB; a separately sealed cell, three graded seeds) shows the target gap is a structural property of AMC, not of one SNR (Fig. 1): because the certificate always selects the MCS that rides the eMBB target, the URLLC consumer reads a false-clear of 0.08–0.14 ($77\text{--}142\times$ its target) at *every* in-range SNR (9–18 dB), and still $54\times$ at 21 dB where the MCS set saturates. The target-aware certificate holds URLLC across the range. At cell edge the relativity compounds: below ≈ 6 dB even the lowest MCS cannot meet the eMBB target and the certificate false-clears *its own* consumer ($\text{BLER} \approx 0.33$ at 3 dB against 10^{-1})—margin cannot fix a certificate whose weakest format is already too fast.

IV. THE REFRESH HORIZON

The second axis is time. A CSI report is fresh when issued and ages as the channel decorrelates; a certificate that was sound at report time false-clears once the channel has moved. Here we use the empirical HARQ NACK rate (2). At $f_D = 200$ Hz ($v \approx 62$ km/h at 3.5 GHz) with a 20-TTI report period, the aged estimate has a mean absolute error $\frac{1}{T} \sum_t |\gamma(t) - \hat{\gamma}(t)| \approx 5.7$ dB (aging-dominated; report noise contributes ~ 1 dB), and the naive (held-CSI) certificate reaches $\text{BLER} \approx 0.36$ against a 0.1 target—a false-clear on roughly a quarter of transmissions—while OLLA (3), the deployed HARQ-witnessed correction, holds it at ≈ 0.10 (Fig. 3(a) shows the aging across mobility).

The operational question is how often to refresh. We sweep Doppler $f_D \in \{10, 25, 50, 100, 200, 400\}$ Hz (T_{coh} from 42 down to 1 ms, an order of magnitude) and report period $P \in \{1, \dots, 80\}$ TTI, define the *refresh floor* $P^*(f_D)$ as the largest P keeping BLER at target, and fit it by least-squares *through*

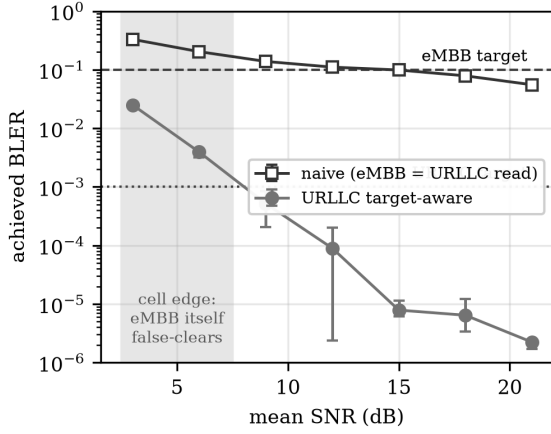


Fig. 1. SNR-robustness of the target relativity (real NR, Sionna LDPC; sealed sweep, 3 seeds, whiskers min-max). The naive certificate rides the eMBB target, so the URLLC consumer reads a $\sim 10^2 \times$ false-clear at every in-range SNR; below ≈ 6 dB (shaded) even the eMBB consumer is false-cleared. The target-aware certificate holds URLLC throughout.

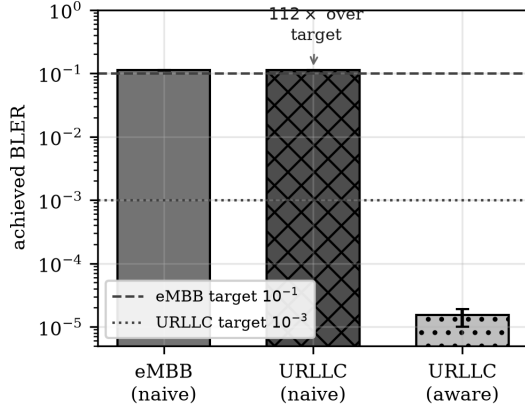


Fig. 2. Consumer-relativity of the CSI certificate in the reliability target (real NR, Sionna LDPC; 3 sealed seeds, $T=6000$ TTI each; bars are means, whiskers min-max). The identical CSI-derived MCS meets the eMBB target (10^{-1}) but false-clears the URLLC target (10^{-3}) by $112\times$; a target-aware policy (conservative MCS + $K=3$ independent diversity branches) meets URLLC.

the origin (linear, not log-log) against the Clarke coherence time $T_{\text{coh}} = 0.423/f_D$:

$$P^* \approx 0.177 T_{\text{coh}}, \quad R^2 = 0.915 \quad (4)$$

(Fig. 3(b)). The fit is dominated by the low-mobility points; at high Doppler the floor saturates at the minimum report period (1 TTI, the dotted line in Fig. 3(b)). This saturation is operationally decisive: above a mobility threshold ($f_D \gtrsim 200$ Hz here) no reporting cadence the frame permits keeps the naive certificate at target, so a HARQ-witnessed correction (OLLA) stops being optional and becomes mandatory. Below it, (4) is a concrete provisioning rule: re-report before the channel decorrelates past $\approx 0.18 T_{\text{coh}}$, not on a fixed calendar.

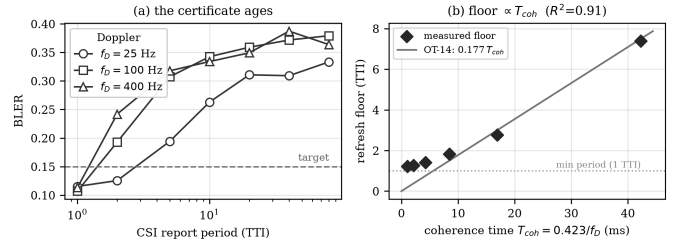


Fig. 3. The CSI certificate's refresh horizon (real NR, Sionna LDPC; sealed sweep). (a) The certificate ages: BLER grows with report period, faster at higher Doppler. (b) The refresh floor is proportional to the Clarke coherence time ($0.177 T_{\text{coh}}$, $R^2 = 0.915$); at high Doppler it saturates at one TTI.

TABLE II
SEALED PER-SEED RESULTS (ACHIEVED BLER)

(a) Reliability target (§III), fresh CSI				
seed	eMBB	URLLC-naive	URLLC-aware	
20260824	0.1119	0.1119	1.7×10^{-5}	
20260825	0.1103	0.1103	1.9×10^{-5}	
20260826	0.1125	0.1125	$< 10^{-5}$	
(b) Refresh horizon (§IV), f_D =200 Hz, 20-TTI				
seed	naive	OLLA	fresh	CQI err
20260823	0.366	0.101	0.116	5.61 dB
20260824	0.340	0.102	0.115	5.86 dB
20260825	0.373	0.101	0.113	5.76 dB

Seed IDs are campaign date-stamps. (a) expected rate; (b) empirical NACK rate.

V. RELATED WORK

Link adaptation, CSI aging, and outer-loop correction are mature. OLLA [1] adjusts an SINR offset from HARQ ACK/NACK to hold a BLER target under imperfect or stale CSI; it is precisely the witnessed correction our refresh-horizon result quantifies. Short-packet and URLLC analysis [3] establishes the reliability-target regime in which the eMBB-to-URLLC gap of §III lives. Current practice configures CQI reporting on a fixed periodicity [4]; our refresh floor recasts that period as a coherence-time budget ($\approx 0.18 T_{\text{coh}}$)—a mobility-adaptive rule rather than a fixed calendar. Learned CSI feedback trained on reconstruction error [6] is an active direction; we note only that the reconstruction-versus-consumer objective gap it might imply did not appear at practical operating points in our controlled tests, and we make no claim on it here. Our contribution is orthogonal: not a better estimator, scheduler, or offset rule, but a *measurement*—the false-clear rate as a first-class, consumer-relative KPI—and the two relativities and the refresh law it exposes.

VI. DISCUSSION AND CONCLUSION

The unit of correctness in link adaptation is not the CSI report but the pair (report, consumer). Sizing margin and refresh cadence for the consumer's reliability target and coherence time, rather than for a nominal channel, is what keeps a certificate honest. The false-clear rate makes the cost of ignoring the consumer measurable: $112\times$ for a mismatched target, a coherence-scaled horizon for a stale report. The Sionna link-level chain is the evidence rung; an over-the-air or

srsRAN testbed with real HARQ traces is the external-validity graduation.

Scope. The evidence is one TDL-A profile (DS 30 ns), an MCS set to 64-QAM, and—for the target axis—one mobility; the refresh law is fit across $f_D = 10\text{--}400$ Hz and the target relativity is measured across mean SNR 3–21 dB (Fig. 1), including the cell-edge regime where the naive certificate false-clears even its own eMBB consumer. The remaining omitted factors point the same way: 256-QAM adds steeper cliffs that deepen the aging penalty, and richer profiles or correlated diversity raise FC_{aware} —each increases, not removes, the measured cost. On realization: the measured refresh floor is directly implementable by the event-triggered (relative-threshold) CSI reporting already in NR [4], an instance of send-on-delta (Lebesgue) sampling [5]; in our exploratory runs such a trigger reproduced the floor constant ($P_{\text{eff}} \approx 0.2 T_{\text{coh}}$), but a sealed confirmation attempt returned VOID on a control-sanity check at very low Doppler, so we make no sealed claim and credit the mechanism entirely to prior art. The methodology is not radio-specific—the same consumer-relative false-clear appears in optical quality-of-transmission and database replication [7]—but the results here stand on the 5G NR CSI certificate alone.

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